



EN-ROADS CLIMATE SIMULATOR ANALYSIS

Policies Proposal

Abstract

There is no denying the fact that the approaching threat of climate change is imminent. In Habib University's 6th Yohsin Lecture, The renowned linguist, philosopher and social critic, Professor Noam Chomsky stated it to be one of the four things that will lead to a terminal disaster and the end of humanity. The paper will look into certain set of policies to counter the threat of severe climate change using data and projections from the En-Roads climate change simulator by Climate Interactive and MIT Sloan Sustainability Initiative. Furthermore, it will also take a look into the extent of its Goals, realism, the extent of equity, implications, and possible failures.

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En-Roads Climate Simulator Analysis

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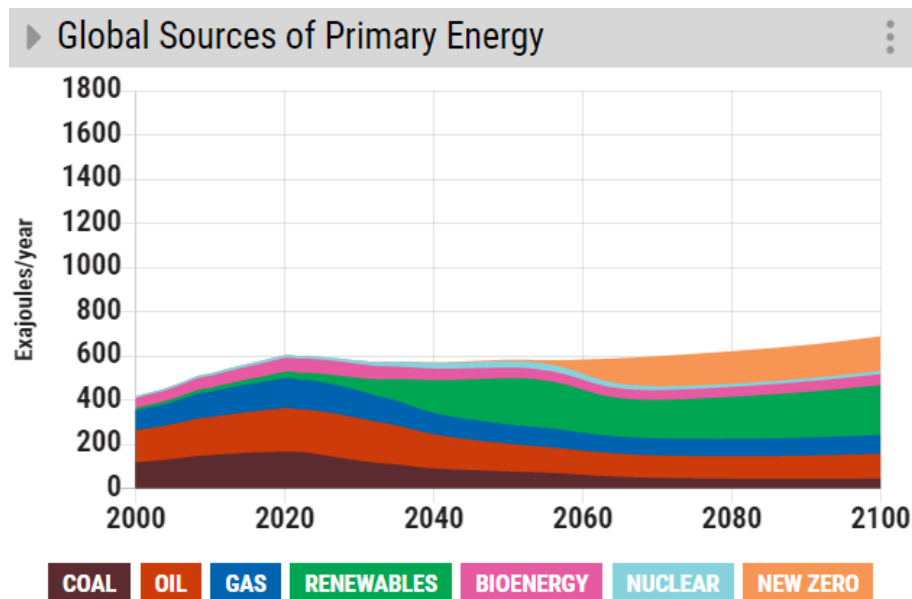
ENVS-102: Introduction to Environmental Systems

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The Plan

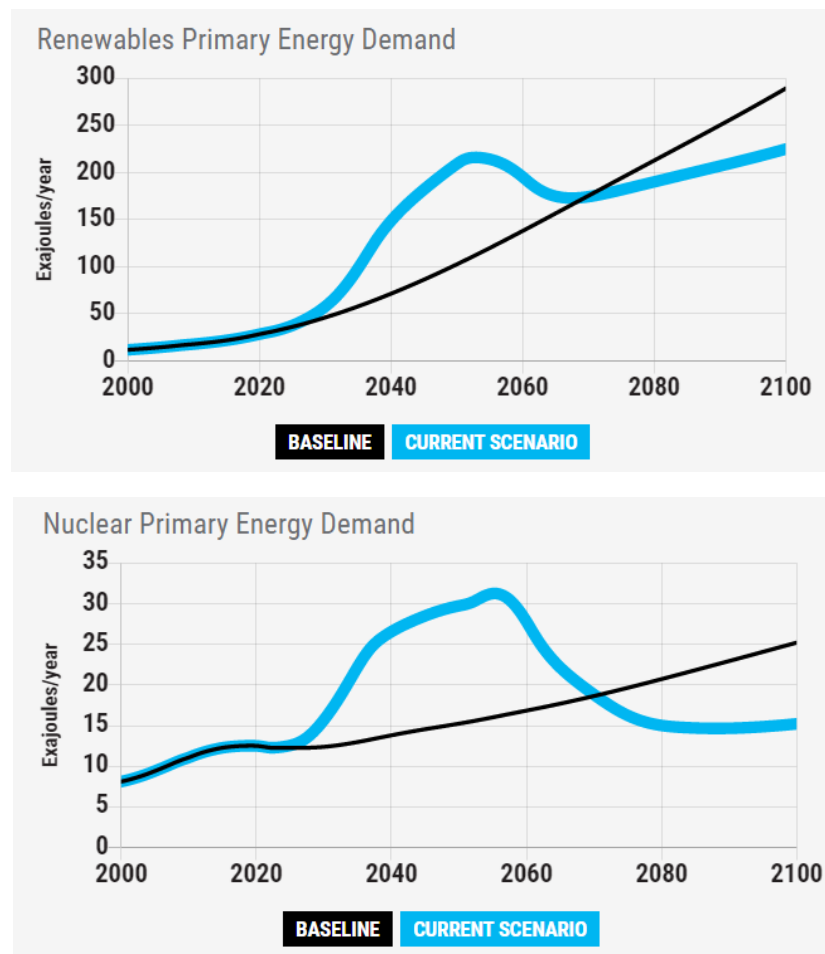
In order to gain a better insight into the effects of several government policies on Climate Change, and more specifically, on an increase in the average global temperature, ‘En-Roads Climate Simulator’ by Climate Interactive and MIT Sloan Sustainability Initiative was used to achieve the minimum temperature of 2 °C [3.6°F] in simulation. The 2015 Paris climate agreement, ratified by nearly all nations, called for “holding the increase in the global average temperature to well below 2 °C [3.6°F] above pre-industrial levels and pursuing efforts to limit the temperature increase to 1.5 °C [2.7°F].” Global warming above 2°C [3.6°F] will cause impacts that will harm our prosperity, security, health and lives. Limiting warming to less than 2°C [3.6°F] is not safe either as we are already seeing impacts today. However, it is a benchmark by which scientists and world leaders believe we can lower the expected harms and risks of catastrophic damage to a manageable level.



Polices Proposal

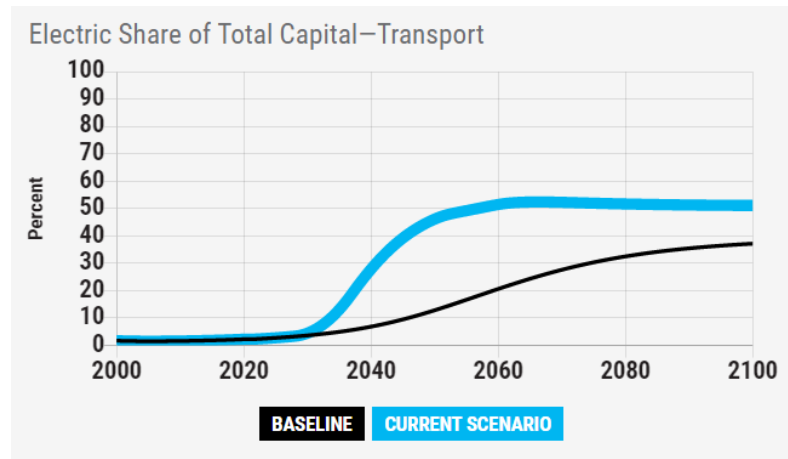
The following set of policies is deemed suitable to achieve that manageable level as per the results of running the simulator. The first measure would be highly taxing the coal usage and mining by around USD 30/tce. By the end of this century, the target of 1/4th of current coal usage will be achieved. Global Oil Shortages are already causing huge fluctuations in oil prices, and disincentivizing oil usage too much would render millions of fuel cars useless as most people would not be able to afford the exorbitant price of petroleum. However, taxing oil is important since it is also one of the key sources of our current carbon emissions. Taxing Oil at USD 20/boe would work fine for most parts of the world. While Coal is definitely the most harmful fossil fuel, and Oil

damage the ecosystems and water, taxing the use of natural gas would prove almost as difficult. Despite having lesser carbon dioxide emissions than coal and oil, methane can affect air and water quality. Hence the optimal rate for discouraging natural gas while being mindful of its use as a cooking fuel would be USD 0.9/Mcf. In addition to disincentivizing the aforementioned non-renewable sources, the carbon price of USD 90/ton CO₂ could be set over the period of the next 12 years. Having discouraged the prominent sources of energy that we use right now, alternative sources of energy that are encouraged are essential to meet the demands of electricity and energy. Subsidizing renewable sources of energy, and nuclear energy both at the subsidy of USD 0.02/kWh would do that. Solar Panels and Wind Farms could work for countries with even no rivers.



Nuclear Energy could be used as a temporary source as we see the world transition to renewable energy fully. The nuclear plants will have a life and could be decommissioned once that is over. Highly skilled personnel for managing its operations should be hired. A nuclear power plant should be built in a remote area in order to minimize the effects of a meltdown. Moreover, with the expected advent of Nuclear Fusion Technology this century, it will be a great addition to our zero-carbon energy sources. This also comes with the lowest external cost of all energy resources we have so far (Entler et al., 2018). For the scope of this paper, the assumption will be that Nuclear Fusion Breakthrough will be in 2030 with 20 years for commercialisation.

Next is probably the most deciding factor, increasing the energy efficiency of our homes, transports and electrical appliances through research in engineering disciplines and by innovating new technologies. We every year see technology that is more efficient than the previous. Advent of LED light bulbs and its replacement with traditional bulbs like tungsten filament bulbs in Pakistan is one such example. Increasing efficiencies will allow us to use less energy and hence reduce our impact on the environment. Another of the latest technologies would be the rise of electric vehicles which not only are more efficient but more economical for the customer. Governments around the world are already subsidizing electric vehicles. Keeping in mind the time lag of implementation, this paper will assume that the acceleration in the electrification of road and rail transports to accelerate in 2027 with the goal of 60% Electrification. Any more than this would be not plausible because of rural populations in underdeveloped nations. This will reduce carbon emissions greatly.



Reducing deforestation which is not that strong factor compared to others above is still relevant in this discourse because trees have many other benefits as well such as reduction in soil erosion. The goal of this environmental policy would be to reduce it by 3.5% per year and dedicate 45% of land for afforestation. 10% of the world's land is considered remote and 95% of the population is concentrated in the other 10% of the land (European Commission). This allows for huge areas with afforestation potential. Another target is to reduce emissions of non-CO₂ gases such as methane, nitrous oxide, and F-gases to 50% by the reduction in the use of meat and fertilizers. The last goal would be technologically reducing the carbon in the air by 47% of the max potential of CDR technologies.

Meeting the Goals

Climate

If the aforementioned policies are implemented, global temperature will increase by 2 °C and not below that. Going below is the goal of the Paris Agreement 2015. Achieving these goals is the bare minimum that would be needed for the survival of the masses and humanity. Any further reduction would prove realistically difficult. However, the solutions stated in the policies above could solve the overarching problem of climate change, but it will create many other problems which would need to be solved individually.

Environment

One of them is the nuclear waste that would be produced thanks to the subsidy being provided to nuclear power plants. Sending nuclear waste to space would be one of

the solutions on the expensive side. The second expensive would be deep-geological disposal in which the waste is disposed of between 250-5000m below ground. “Deep groundwater is devoid of oxygen, minimising the possibility of chemical mobilization of waste” (World Nuclear Association, 2021). The cheapest option would be to build a large on-ground isolated facility for storing waste, and this could work for developing countries.

An increase in the number of hydroelectric power plants will reduce the amount of water in the river and might destroy the delta of that area as the life of the delta is dependent on the availability of freshwater. This might endanger many marine species, and reduce the local fish population. An increase in wind farms will require more land, and bird populations might be endangered in certain areas. Therefore, it would be crucial to research areas, collect data on the number of birds, and their migration patterns, and set up relevant monitoring systems. “The current challenge is to develop suitable techniques for offshore assessments. The application and further development of ship-based and aerial survey techniques have been invaluable in carrying out assessments. The next step is to develop and successfully apply remote data-gathering equipment such as radar and infra-red cameras to provide information on bird distribution and movements” (Drewitt & Langston, 2006). However, the cheaper alternative would be to use the research conducted to decide where to set up the wind farms so that the death rates of birds are minimised and the ecosystem of that area is not compromised in any form.

As far as solar panels are concerned, mass-producing them may require the use of toxic metals such as tin in the production of Photovoltaic cells made from perovskite materials. This is where the possibility of materials such as tin needs to be further explored for disadvantages (Peplow, 2014). Apart from that, proper hazard and safety rules must be created for workers so that harmful gases do not affect any lives or the environment. Recycling ought to be the best viable option for disposing of the solar panels after their life ends which is up to 30 years. Some of the many toxic chemicals produced are Arsenic, Selenium, and Cadmium. It is possible that some of these chemicals may no longer be used for PV devices production as the fabrication processes are constantly changing and evolving (Nkuisi, 2020). Many recycling techniques exist. “Some major PV manufacturers have experienced a promising approach called “cradle to cradle”. The concept of this approach is to recycle the toxic materials of the process into new products, which are less or not at all toxic” (Nkuisi, 2020). Along with that many methods for recycling exist such as chemical stripping, electrodeposition, precipitation, and evaporation, which recycles the panels (Nkuisi, 2020). All in all, the real problem lies with the commercialization of such recycling facilities by making sure that these facilities exist in every corner of the world be it an underdeveloped, developing or developed country. In one of the International Renewable Energy Agency’s reports, it was recommended to expand waste management infrastructure as well (Barnes, 2017).

Equity

The extent of equity is somewhat on the negative side. It might not be possible to achieve 100% equity but these policies will definitely produce an outcome that is somewhat less bad. Coal taxation will not decommission coal miners, because coal will

still be a valuable commodity in the production of metals, chemicals, and cement. Employment at fuel stations will reduce significantly as electric vehicles become common. As far as other jobs are concerned, the difference in policies will create new jobs and reduce the number of positions in previous jobs. Ultimately, this will change the very job preferences that exist in our current society and will create further economic classes. Developed countries might find it feasible to adopt these policies, but others might find it difficult to import Uranium or other resources that they do not have locally. This adds extra strain on their state bank reserves. But seeing the trend of demographic transitions throughout the world, it is safe to assume that these countries will too.

Secondly, even if some countries fail to adopt the measures, the other developed countries might outperform their goals and fulfil the requirements on behalf of the other countries. They might do this to achieve better political standing, trade, or collaborations. Tribal Areas would continue deforestation and use bio-fuels or firewood, but due to their small population, the effect of it would be negligible. The main impacts come from Urban due to their relatively higher population size and density which is why it is important that these policies are enforced more in urban areas instead of rural. Moreover, it is important to note that with growing globalisation and urbanisation, people tend to move to cities to find jobs and to improve their overall quality of life. This means that rural populations will continue to decline, while the urban populations will continue to increase as we approach the end of this century. “Globally, over 50% of the population lives in urban areas today. By 2045, the world's urban population will increase by 1.5 times to 6 billion. City leaders must move quickly to plan for growth and provide the basic services, infrastructure, and affordable housing their expanding populations need” (World Bank). This gives another reason to enforce these policies in urban areas despite the lack of somewhat equality. As far as electrification is concerned, people might feel it difficult to stop using natural gas, firewood or coal for cooking, despite the benefits that induction cooker offers such as faster and better cooking. Awareness of induction cookers will be the key to convincing the public of switching to them. Moreover, by taxing natural gas, the public will be forced to look for alternatives. The same goes for electric cars, which have many electric charging stations in western countries.

On the contrary, let us take Pakistan for example where despite incentives for electric vehicles, they are not famous amongst people mainly due to the lack of charging stations. Karachi alone contains only two commercial charging stations, and it would take around 2 hours on average to charge one vehicle. This shows that Electric Charging stations need to be even more in number than the existing petrol pumps. In order to be mindful that people will stay for hours at charging, it is imperative that they are available at sources of entertainment or food such as shopping malls, and restaurants. This will convince even the less educated class to switch to an electric motorcycle and not a petrol one. Reducing methane and non-CO₂ emissions depower the working classes of agriculture and the whole beef industry. This might be achievable again if people be rational through sufficient education and understand the adverse impacts of supporting meat. Hence, there will be a global imbalance in the consumption of beef. Awareness campaigns could play a vital role in making the public aware and hence

reduce the demand for meat in the market, which will ultimately cause the suppliers to reduce the supply. It can be reasonably inferred that awareness will only remove a very small percentage of inequity.

Economy

Sudden implantation of these policies might be difficult, as there might be a global shortage of engineers and technicians. This means that in the future the job market will be engineering-oriented more. Some countries might take a loan to implement these policies right away. To predict whether they will be able to get out of a 'debt trap' is difficult. However, due to trends we have seen in developed countries, we saw that as their economic activity and economic growth increase their population growth decreases. Let us take a closer look at this. Advancements in medical technologies and nutrition lead to a demographic transition. Ideally, this would mean low birth rates, low death rates, and hence low population growth or even depopulation in some cases. The reason for high-birth rates usually is a lack of education, the need for a workforce, and the lack of working women. An example would be that in an underdeveloped country, a farmer would need children to support his work as manual labour. A working woman might prefer her work instead of increasing her family size. However, these reasons are eliminated due to social, technological, and medical reforms in a country. This causes a change in population dynamics which we call a demographic transition of a country. Falling death rates are one of the signs of such transition hence it could be safe to assume that other reforms might follow, which will also cause a reduction in total population growth, low birth rates, and low fertility rates. A declining infant mortality rate would imply declining fertility rates in the near future as well. This phenomenon is called a demographic transition of a country. "The demographic transition has enabled economies to convert a larger portion of the gains from factor accumulation and technological progress into the growth of income per capita. It enhanced labour productivity and the growth process via three channels" (Galor, 2012). If the whole world were to achieve such a demographic transition this would mean that the world would soon have a greater economy. The new sources of electricity which are renewables, and nuclear have fewer external costs. So countries which produce surplus electricity and sell it to other countries will be more dominant in terms of economy.

Realism without cynicism

The aforementioned proposal does contain policies that make certain assumptions that might not turn out to be true. One of those predictions is that nuclear fusion reactor will be in working conditions by 2030 with 20 years for commercialisation. How do we make sure we meet these deadlines? For that, we need to have really good PR for Fusion reactors. "How long it will take for fusion energy to be successfully rolled out will depend on mobilizing resources through global partnerships and collaboration, and on how fast the industry will be able to develop, validate and qualify emerging fusion technologies" (Barbarina, 2022). Manufacturing a huge amount of commercial charging stations will take a lot of time, but will convince the population to switch to Electric vehicles. A lot of petrol pumps will have to be decommissioned, and this will waste a lot of construction resources that were used to make those. Implementation of technological carbon-removal solutions through CDR technologies will prove difficult as

these are not yet widely available. So assuming that we will have them at 47% is probably the biggest challenge this proposal faces. This is the most expensive thing that will hinder many of the developing nations. Right now, countries do not implement such technology because of little economic benefit from it. Therefore, using this technology can be saved as a last resort. Awareness in a developed country's government might convince them to allocate funds for this technology, but the story is very different from the rest.

Implications

Winners/Losers and role of businesses

Engineering Companies will receive the most funding, and subsidies while the oil industry and meat industry will suffer a heavy setback. Unemployment of workers in these industries will have them looking for jobs elsewhere which will be difficult due to a huge shift in the job market and its demand. The most difficult position would be for banks as they would have to decide whether to give loans or not. Banks might profit in one region whereas the companies that took a loan in another region might go bankrupt. Construction companies would also benefit as new nuclear power plants, fusion reactors, and hydroelectric plants may be needed for construction. They would also need to develop more energy-efficient housing and buildings so that the energy losses are minimized. The role of business will be simple. To do relevant collaborations, awareness spreading campaigns, optimize their services and manufacturing processes so that they are energy efficient. Most Importantly, they will need to work with governments to make sure policies are implemented swiftly. Tech companies will need to support the engineering companies, for instance in developing monitoring systems that could be used to collect data on birds. This way wind farms can be constructed in a location with the least number of losses. Tech businesses could also help by automating many of the operations in the power plants so that chances of human error are minimised. This would make sure we do not see a nuclear meltdown.

Getting Started

Mass production of commercial electric charging stations and construction of nuclear power plants and wind farms would be the focus for the first two years. This would ensure that people buy electric vehicles that are already available and hence the carbon emissions will be reduced significantly. Industries in collaboration with universities should research more on Nuclear Fusion more so that a working reactor is achieved. As electricity production increases through renewables, taxing coal should start, followed by oil. Keeping the electricity production in check. Nuclear Power plants will not be done within the first two years as they take around 5 years to be completed. Awareness campaigns for reducing deforestation, induction cookers, electric vehicles, population control and reducing meat usage should be common by the end of these two years. Engineering students should be given scholarships so that the students are encouraged to take part in this discipline and future demand for engineers is met accordingly. Research on Carbon-Removal Technologies and Energy efficiency of appliances and buildings should be incentivized. This will make sure that future technologies are more energy efficient. Lastly, banks will play a significant role in

coming up with good schemes and loans for construction companies, and contractors to help them achieve these goals. World Bank will continue to give aid to developing countries.

Reflections and Feedback

The sheer magnitude of effect that coal usage for electricity production and energy efficiency can have on climate change is phenomenal. Another surprising takeaway from this discourse would be that reducing deforestation and implementing afforestation alone is nowhere close to solving the problem of climate change. The strongest reduction can be caused by Technological-Removal of Carbon from the air. This would be an expensive operation to run and commercialise. Adverse impacts of wind farms and solar farms are also not very well known. Bad PR of nuclear energy is probably the reason why it is not the most commonly used power plant. All in all, the number of changes in many segments of our society that must be achieved limits the possible sets of policies we can have. Our best chance would be to act as quickly as possible.

Model flaws and areas for improvement

It is quite axiomatic that it is impossible to include all the countless known and unknown environmental factors into the simulator. This did not take into account the ocean and soil health for instance which could play an enormous role in the removal of carbon from the air. The simulator tends to show little change on a decrease in deforestation and increase in afforestation which seems quite inaccurate.

Hope

According to the assumptions and the data that is used in the simulator, it would prove very difficult to implement policies, especially those that are expensive and those in developing countries. The most important thing for society is to understand the gravity of the situation and awareness is the only method through which this could be achieved. It is up to us to spread awareness as much as possible while being optimistic. Planting a tree, reducing meat usage, reducing water usage, reducing electricity usage, and recycling materials is the thing that we can do to make a change on a personal level. It is quite known that for a positive change to come in society, before making policies, we must make sure that the change comes from within us.

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